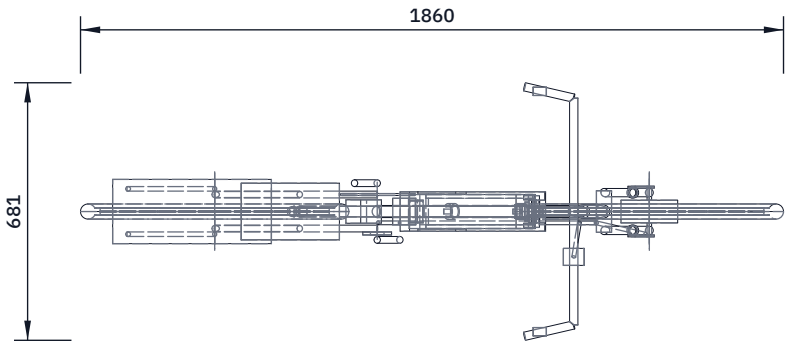


CONCEPT, NOT FOR FABRICATION

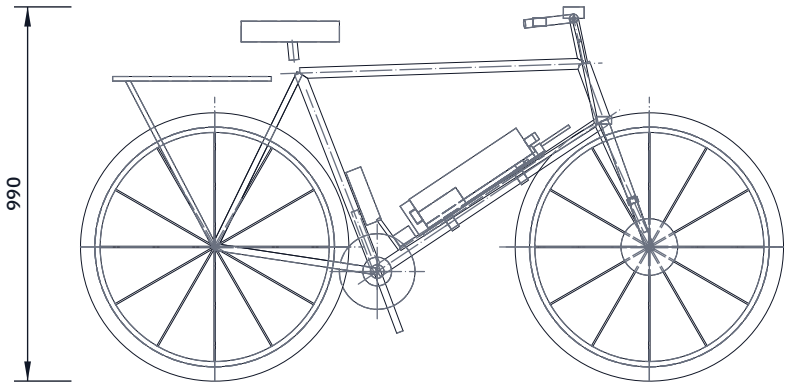
SunSpoke · github.com/BoujeeEnjinia1701/sunspoke

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA, SwapCell interface v0.3 (SSP-CAL-001)	2026-09-25	AC
P2	Notes: motor thermistor and derate (SSP-DDR-002)	2026-09-25	AC



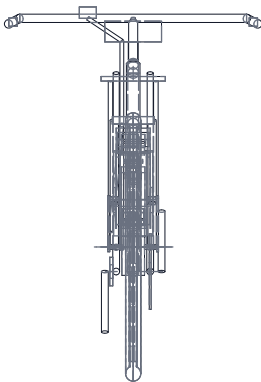
TOP VIEW

Scale 1:20; looking down (along -Z)



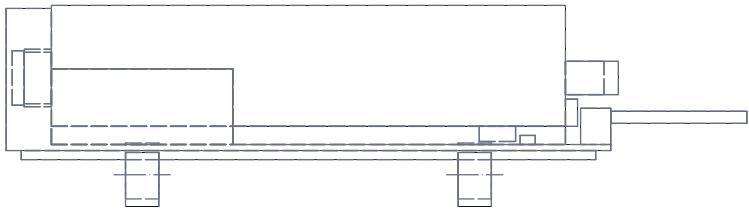
FRONT VIEW

Scale 1:20; looking at the front (along +Y)



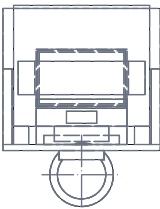
RIGHT VIEW

Scale 1:20; looking at the right side (along -X)



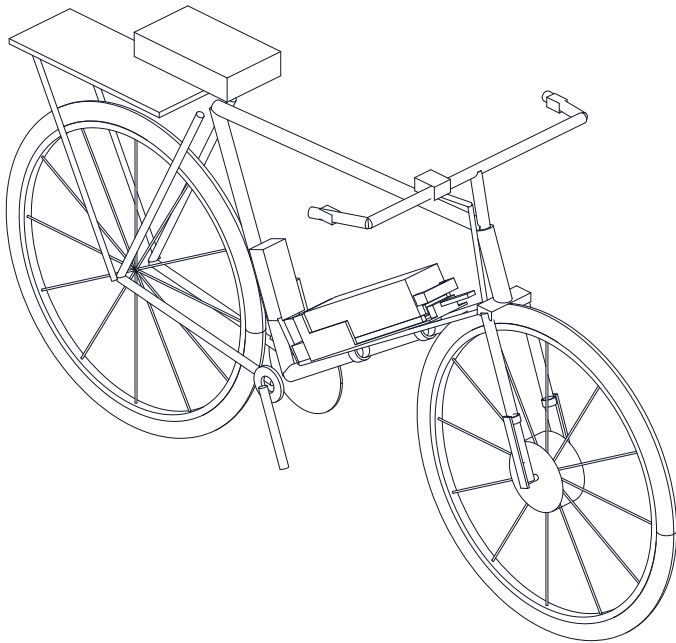
DETAIL A: CRADLE WITH PACK, SIDE VIEW

Scale 1:5; down tube axis horizontal, BB to the left



DETAIL A: END VIEW

Scale 1:5; from the handle end



ISOMETRIC VIEW

Not to scale; seen from the front right and above, about 30 deg elevation

KEY DIMENSIONS AND INTERFACES (MM)

Wheel 28 in (ETRTO 635), tyre radius 355; wheelbase 1150
Motor 1: geared hub, 150 dia, 100 dropout spacing, 10 axle flats
Torque arms 2: 5 x 20 steel, clamp 130 from axle, both sides
Cradle 4: band clamps for 28 to 32 dia down tube, 220 pitch
Pack 6 centre at 48 % of down tube (721 long) from BB
SwapCell v0.3 envelope 340 x 90 x 80, connector toward BB
Guides 120 long, 1 clearance per side
Removal travel 164 free (143 needed); triangle clearance 123
Latch class V1: over-centre lever, 330 N preload, detent
INTERLOCK: 10 kOhm 1 % coding resistor in cradle, power switch 8 in series (interface v0.3 item W)
Controller 3 on seat tube; host adapter 5 below cradle
Motor 1 NTC thermistor to controller 3; derate 110 C
Solar set 10 to 13 not shown (see SSP-DWG-010)
PRELIMINARY, NOT FOR FABRICATION

SUNSPOKE

github.com/BoujeeEnjinia1701/sunspoke

General arrangement, kit on 28 in roadster

DRAWING NO. SSP-DWG-001		REV P2	SHEET 1 of 1	SIZE ANSI B
SCALE 1:20	UNITS mm	PROJECTION 	DATE 2026-09-25	DRAWN Amish Chadha
MATERIAL / NOTES Kit parts per bom/bom.csv; donor roadster shown for context				
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